



Assessment Analysis Guide
25-26 Math G7 Form U Prop Rel & Sci
Mathematics
Grade 7

Reference #: 688005c

1

Item Id: i184987

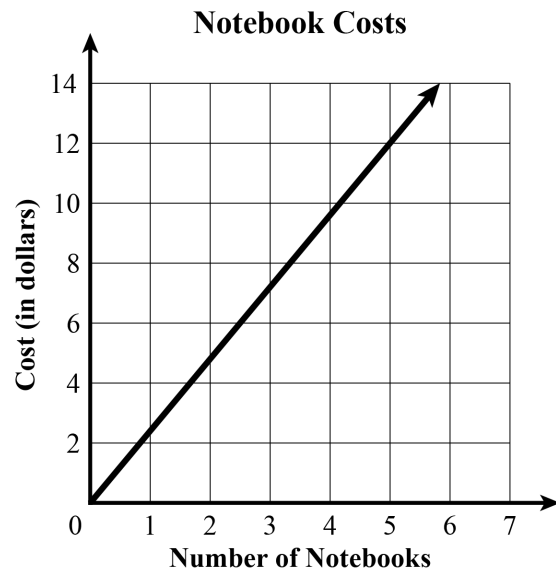
Item Type: SR (Selected Response)

Standards Description:

7.RP.A.2.c Represent proportional relationships by equations.

Content:

The graph below shows the cost, in dollars, per notebook at the school store.



Which equation represents the total cost, t , per n notebooks?

A $n = 2t$

B $n = 2.4t$

C $t = 2.4n$

D $t = 0.42n$

Primary Standard: 7.RP.A.2.c

Points Possible: 1

Distractor Rationale:

A: $n = 2t$

Student used the interval on the vertical axis as the constant of proportionality and switched the variables.

B: $n = 2.4t$

Student switched the variables in the equation.

C: $t = 2.4n$

Correct.

D: $t = 0.42n$

Student used a value from the graph but represented the inverse unit rate, OR switched the variables.

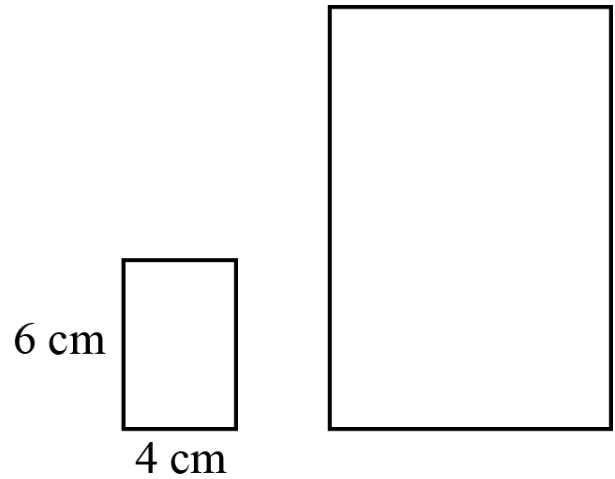
2

Item Id: i184951**Item Type:** SR (Selected Response)**Standards Description:**

7.G.A.1 Solve problems involving scale drawings of geometric figures, including computing actual lengths and areas from a scale drawing and reproducing a scale drawing at a different scale.

Content:

The large rectangle below was created from the small rectangle using a scale factor of 2.5.



What is the area of the large rectangle, in square centimeters?

- | | |
|---|-------|
| A | 26.5 |
| B | 55.25 |
| C | 60 |
| D | 150 |

Primary Standard: 7.G.A.1**Points Possible: 1****Distractor Rationale:**

A: 26.5

Student added the scale factor to the original area rather than multiplying each dimension by the scale factor.

B: 55.25

Student added the scale factor to each dimension before calculating the area, rather than multiplying each dimension by the scale factor.

C: 60

Student applied the scale factor to the area rather than recognizing that the scale factor must be applied to each dimension separately.

D: 150

Correct.

3

Item Id: i184986

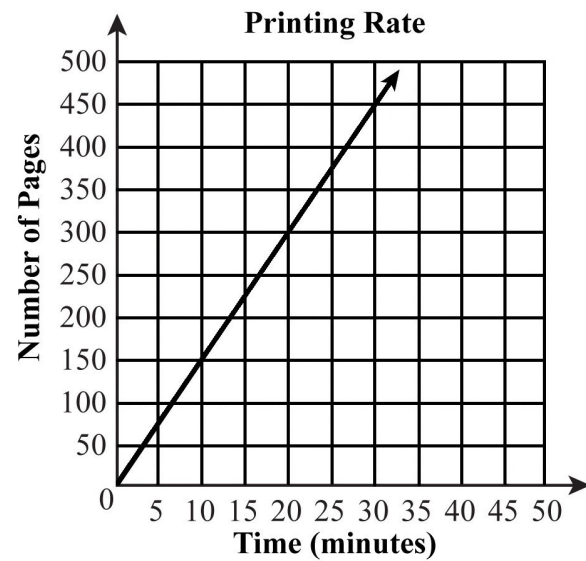
Item Type: SR (Selected Response)

Standards Description:

7.RP.A.2.b Identify the constant of proportionality (unit rate) in tables, graphs, equations, diagrams, and verbal descriptions of proportional relationships.

Content:

The graph below represents the number of pages a printer can print over time.



How many pages does the printer print per minute?

- A 5
- B 10
- C 15
- D 50

Primary Standard: 7.RP.A.2.b**Points Possible: 1****Distractor Rationale:**

A: 5

Student interpreted the first labeled interval on the horizontal axis as the rate of change instead of determining the actual unit rate.

B: 10

Student selected a point from the graph to find the unit rate but used only the horizontal value instead of performing the necessary calculation.

C: 15**Correct.**

D: 50

Student interpreted the first labeled interval on the vertical axis as the rate of change instead of determining the actual unit rate.

4

Item Id: i184985

Item Type: SR (Selected Response)

Standards Description:

7.RP.A.2.c Represent proportional relationships by equations.

Content:

Kenwyn is making flower arrangements for an event. The table shows the number of flower arrangements, f , she can make in m minutes.

Time, in minutes (m)	Number of flower arrangements (f)
32	4
48	6
64	8
80	10

Which equation represents the relationship between the number of flower arrangements she can make and the time, in minutes, it takes her to make the flower arrangements?

A $f = 0.125m$

B $f = 2m$

C $f = 8m$

D $f = m + 16$

Primary Standard: 7.RP.A.2.c

Points Possible: 1

Distractor Rationale:

A: $f = 0.125m$

B: $f = 2m$

C: $f = 8m$

D: $f = m + 16$

Correct.

Student used the change in one quantity as the constant of proportionality instead of determining the correct ratio between the two quantities.

Student set up the equation so that multiplying by the constant would yield the wrong variable, misplacing the values in the relationship.

Student applied an additive pattern based on the interval increase in one quantity instead of identifying the proportional relationship between the two quantities.

5

Item Id: i184949

Item Type: SR (Selected Response)

Standards Description:

7.G.A.1 Solve problems involving scale drawings of geometric figures, including computing actual lengths and areas from a scale drawing and reproducing a scale drawing at a different scale.

Content:

Two students made scale drawings, in inches, of the same road.

- Anna used a scale of 1 : 600, where 1 inch of her drawing represents 600 feet.
- Anna's scale drawing was 6 inches long.
- Greg used a scale of 1 : 300, where 1 inch represents 300 feet.

How long, in inches, was Greg's scale drawing?

A 2

B 3

C 12

D 50

Primary Standard: 7.G.A.1**Points Possible: 1****Distractor Rationale:**

A: 2

Student divided the scale values and did not understand how to use this value to determine the length of the drawing.

B: 3

Student did not first determine the actual length being represented before applying the new scale, leading to incorrect proportional reasoning, OR student thought that halving the scale factor meant halving the corresponding length.

C: 12**Correct.**

D: 50

Student applied division within the given scale rather than using proportional reasoning to adjust the length according to the new scale, possibly confusing the scale factor with the conversion process.

6

Item Id: i184982

Item Type: SR (Selected Response)

Standards Description:

7.RP.A.2.b Identify the constant of proportionality (unit rate) in tables, graphs, equations, diagrams, and verbal descriptions of proportional relationships.

Content:

In the table below, the number of pounds of blueberries is proportional to the number of pounds of strawberries.

Pounds of Strawberries	Pounds of Blueberries
2	$1\frac{1}{2}$
8	6
$10\frac{2}{3}$	8
20	15

What is the number of pounds of blueberries per pound of strawberries?

A

$\frac{1}{2}$

B

$\frac{3}{4}$

C

$1\frac{1}{3}$

D

$1\frac{1}{2}$

Primary Standard: 7.RP.A.2.b

Points Possible: 1

Distractor Rationale:

A: $\frac{1}{2}$

Student determined the difference between the two quantities in the first row instead of finding the ratio.

B: $\frac{3}{4}$

Correct.

C: $1\frac{1}{3}$

Student calculated the unit rate using the inverse relationship, instead of the intended unit rate.

D: $1\frac{1}{2}$

Student selected a ratio from a specific row where both values were whole numbers but misinterpreted the fraction, leading to an incorrect equivalent ratio.